

Mauricio González Ortiz

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BACKGROUND

Backend and AI Engineer with a background in physics and mathematics, specializing in scalable AI systems and production-grade software. I have 3 years of experience building and leading engineering solutions in startup environments, with expertise across architecture, APIs, deployment, and production operations. My work focuses on computer vision, LLM-based agent systems, RAG/ETL pipelines, and performance optimization. I enjoy solving complex problems, improving system efficiency, and building reliable AI infrastructure while collaborating with clients and stakeholders to deliver impactful solutions.

EDUCATION

Tecnológico de Monterrey

B.S. Industrial Physics Engineering • GPA: 97/100 • Best Score of Program

Monterrey, NL

June 2023

Google Google Data Analytics Professional Certificate • SQL, Tableau, R, Excel

July 2023

EXPERIENCE

Parvada

Clovis, NM

Software & Deployment Engineer — AI/ML

Feb 2025 – May 2026

- Eliminated manual chicken classification, processing 3000 chicks/minute, by creating an end-to-end AI pipeline in Python and C++. Using YOLOv8 object detection, wing localization and a ViT for classification with the integration of PCIE actuators, for automatic separation.
- Reduce transportation cost by 50% by developing and deploying a local AI agent (using openai when connection to the internet was available and llama when it wasn't) for system diagnostics, capable of identifying software vs hardware failures, applying software fixes and pinpointing what components needed maintenance.
- Reduce inference latency by 33% by implementing a grayscale inference solution in Ultralytics before native support was available.
- Increase the model accuracy by 60% by creating a high-performance C++ machine vision library for an industrial speed camera system, enabling asynchronous frame acquisition with zero-copy transfers (from camera capture, to pin memory and direct to GPU inference), minimizing data overhead and increasing the amount of frames per chicken.
- Designed and implemented a custom C++ object tracker, capable of re-identifying object IDs 10x faster than conventional tracking algorithms.
- Got a 30% increase in clients by shipping a TypeScript/Node.js mobile app with hybrid CNN inference (running models on-device when offline and calling Amazon SageMaker API endpoint when connected). The app was able to classify chickens by gender and count how many chickens were in an image.
- Delivered 50% reduction in client operational costs by independently owning the full deployment lifecycle (hardware assembly, model training and integration, machine validation, and post-launch monitoring) serving as the sole technical point of contact across three live production systems.

Okus AI

Remote, USA

Data Scientist / ML Engineer

Feb 2024 – Mar 2025

- Double production throughput by designing and implementing an end-to-end ELT pipeline that extracted raw video data from multiple factory areas, processed it using CNN-based pose detection and segmentation models, and converted visual information into structured production records (process name, workers involved, machine ID, start/end times, and other operational metrics).
- Increase the number of clients by 100% by building the data infrastructure to store process records in Amazon DynamoDB, and integrating an LLM agent using Amazon Bedrock to analyze records and create a full report of operation.
- Built a Python data pipeline that ingested 50K+ ClinicalTrials records, applied NLP-based entity extraction and market-sizing models to rank addressable fields with AI adoption potential.

RESEARCH EXPERIENCE

Harvard Medical School – Zhang Lab

Boston, MA

Undergraduate Research Fellow — Software & Computational

Feb 2022 – Feb 2023

- Cut manual microscopy annotation time by ~60% by building Python image processing pipelines that automatically analyze cell distribution, quantifying relationships between cell size and population frequency.
- Implemented finite element simulations for soft magnetic materials using FORTRAN and ABAQUS, modeling material behavior and physical interactions.

INDEPENDENT PROJECTS

ScienceCompanion

2024 – Present

github.com/bebedovis/ScienceCompanion

Multi-Agent RAG System

- Designed a generative AI agent system using LangChain and LangGraph, where given a query the agent retrieves documents from ArXiv API or downloaded papers, and creates a full context driven response, with correct citation. Chunking is made by sections and use of regular expressions, retrieval is a hybrid method using BM25 index and a vector store (before in ChromaDB and now in AWS OpenSearch). LLM agents can be switched between openai and Ollama.

MauriGPT

2023 – Present

github.com/bebedovis/MauriGPT

Custom Transformer & Fine-tuned LLM

- Created and deployed a custom LLM using QA training method and WhatsApp chat history as training data. It has 2 LLMs, one built from scratch and the other one using Huggingface Transformers for fine-tuning. The chat is deployed through a FastAPI and frontend through a streamlit App.

BasketBallNoFoul

2023 – Present

Computer Vision for Automated Foul Detection

- Built an automated basketball foul-calling system using YOLOv8 pose estimation to track player keypoints in game footage. Trained a classifier on pre-labeled foul data, using joint-proximity thresholds and pose overlap between players to flag illegal contact.

SKILLS

AI & LLM: LangChain, LangGraph, RAG, Multi-Agent Systems, Generative AI, Prompt Engineering & Version Management, Tool Calling, OpenAI SDK, Anthropic SDK, Ollama, HuggingFace

ML & CV: PyTorch, YOLOv8 / Ultralytics, Computer Vision, Deep Learning, NLP, AI Model Training & Fine-tuning

Evaluation & Monitoring: precision@k, MRR, BLEU, perplexity, model drift detection, retrieval evaluation, regression testing

Vector DBs & Retrieval: OpenSearch (AWS), ChromaDB, Pinecone (familiar), vector indexing, structured extraction, chunking strategies

Backend & APIs: Python, FastAPI, TypeScript, Node.js, C++, Bash, SQL, REST / Webhook design, Git, CI/CD pipelines, pytest

Infrastructure: Linux Systems Administration, Docker, ONNX, Distributed Systems, AWS (EC2, S3, cloud endpoints), Git

Human Languages: Spanish (Native), English (Bilingual)